

various topics on the Internet related to their subjects.

Upgraded learning. With the use of smart technology and electronic gadgets, information can be shared more effectively with the use of photos, maps, graphs, flowcharts, and animated videos. It makes teaching sessions and learning more interactive, easy to understand, and fascinating. Topics can be understood much better through visuals rather than just reading from the blackboard.

Time-saving. This is another major benefit of smart education. In the conventional methods of studies, students need to prepare long hand-written notes, whereas smart education aids students to make presentations online and get feedback from their teachers in much lesser time. The same goes for teachers as well as they do not need to worry about giving verbal or written notes because presentations can be shared directly with all the students.

Conclusion. Overall, understanding

of technology offers enhanced skills required for a digital future. Teachers, on the other hand, are also required to learn new technologies and methods of executing them in conventional classrooms with the utilisation of ICT. By implementing sophisticated learning techniques, such as online virtual classrooms, virtual learning environments, cloud servers, smartphones, etc, teachers can help students to grasp more out of learning.

—Nijhum Rudra

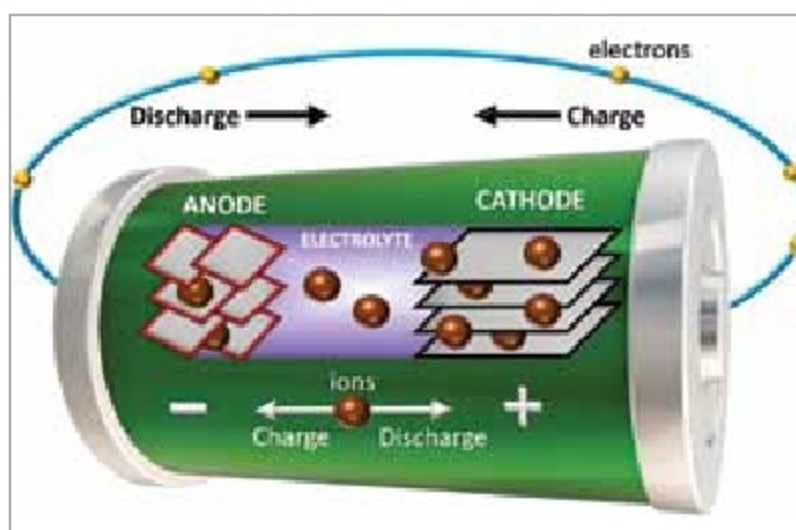
3 BATTERIES FOR POWERING SUSTAINABLE DEVELOPMENT

Over the years, lithium-ion (li-ion) rechargeable batteries have become immensely popular, but their production process incurs large carbon footprints. Hence, there is an urgent need to find alternatives to these batteries

Batteries are a core technology that power portable electronic devices like smartphones, vehicles, and even provide a backup for the growing renewable energy used in homes, offices and electricity grids by converting chemical energy inside them to electrical energy. In the past few years, lithium-ion (li-ion) rechargeable batteries have become immensely popular due to their high energy density over a wide temperature range and low self-discharge, finding their use even in military and aerospace applications.

Just like any other battery, li-ion batteries are made up of a combination of cells, with each cell consisting of anode and cathode that are separated by an electrolyte. The cathode is made up of a lithium compound (such as lithium cobalt), and the anode is typically built from graphite. The electrolyte is usually a lithium salt (lithium hexafluorophosphate) in an organic solvent. During discharge, lithium ions move from cathode through the electrolyte to anode and vice versa during charging. The cells need a protective circuit to limit the peak voltage.

Owing to climate change, we need the development of a sustainable and low-carbon value chain for batteries. The storage of energy heavily relies on heavy metals in batteries. The



Lithium-ion batteries (Credit: www.sigmaaldrich.com)

problem with li-ion batteries is that besides being expensive, extraction of raw materials available in underdeveloped nations is often done in unsafe working conditions, and a huge amount of water gets wasted in the process. Sometimes, it even leads to leakage of toxic chemicals into the water cycle. The production process incurs large carbon footprints, and their disposal is a huge problem. Besides, with China being dominant in li-ion battery manufacturing and battery supply chain, the coronavirus pandemic has led to a series of disruptions in the supply chain.

As the basis for future mobility, there needs to be a defined sustainability criterion for battery design as well as focus on reuse and recycling to boost a circular economy, especially in developing and underdeveloped

countries. But the cost of recycling li-ion batteries is higher than manufacturing these from scratch. For moving to a more sustainable future, these issues need to be fixed. There is a dire need for implementation of effective coordination and policies among public, private, and civil society organisations for the deployment of sustainable batteries. In this direction, the Global Battery Alliance was founded in 2017 for global public-private partnerships.

Several researches are also going on to find alternatives to li-ion batteries. For example, sulphur has been considered to have great potential in this field. Lithium-sulphur batteries are economical, more environmentally sustainable to produce, and safer to use. The issue is that the electrodes tend to break down during charging since sulphur is a poor conductor of electricity and heat. Apart from this, these batteries degrade faster over time.

In another instance, researchers at Purdue University have created a technique to turn waste polyethylene terephthalate, a recyclable polymer, into battery components. This paves the way for organic batteries. Biomass-derived electrodes (both carbonised and non-carbonised), electrolytes and separators (membrane between cathode and anode) are a growing topic of